

Deep Learning and Its Application

Week 10: Practicals

E1.1 Labwork:

- 1) Read the OpenAI Gym introduction
https://gymnasium.farama.org/introduction/basic_usage/
- 2) Install the following packages under Terminal within the PyCharm

```
pip install gymnasium
pip install "shimmy[gym-v21]"
pip install "shimmy[gym-v26]"
pip install swig
pip install "gymnasium[box2d]"
```

- 3) run the CartPole code below

```
import gymnasium as gym
env = gym.make('CartPole-v1', render_mode="human")
#env = gym.make('CartPole-v0')
for i_episode in range(20):
    observation = env.reset()
    for t in range(100):
        env.render()
        print(observation)
        action = env.action_space.sample()
        observation, reward, done, info = env.step(action)
        if done:
            print("Episode finished after {} timesteps".format(t+1))
            break
env.close()
```

- 4) print out “observation, reward, action” and try to understand them
- 5) run the “MountainCar” code below

```
import gymnasium as gym
env = gym.make('MountainCar-v0', render_mode="human")
for i_episode in range(20):
    observation = env.reset()
    for t in range(100):
        env.render()
        print(observation)
        action = env.action_space.sample()
        observation, reward, done, info = env.step(action)
        if done:
            print("Episode finished after {} timesteps".format(t+1))
            break
env.close()
```

- 6) print out “observation, reward, action” and try to understand them
- 7) change time-steps and episodes in the above code and understand
- 8) understand action and observation spaces and print respective information
- 9) run the “LunarLander”

```
import gymnasium as gym
env = gym.make("LunarLander-v3", render_mode="human")
#env = gym.make('MountainCar-v0')
observation, info = env.reset(seed=42)
for _ in range(1000):
    action = env.action_space.sample() # this is where you would
```

```
insert your policy
    observation, reward, terminated, truncated, info =
env.step(action)

    if terminated or truncated:
        observation, info = env.reset()

env.close()
```

E1.2 define in your own words: (a) observation (b) action (c) reward (d) Markov decision process (e) sequence of states (f) episodes and their impact on rewards (g) value function